Unit - 4 Algebraic Structures

(1) **Binary Operation :**
Let G be any non empty set Then $G \times G = \{(a,b) \mid a \in G, b \in G\}$ If $f: G \to G$ then f is said to be binary operation.

- '+' is a binary operation on N $a \in N, b \in N, a+b \in N$
- (−) is not a binary operation on N because $a=1, b=2$
  $a-b = 1-2 = -1 \notin N$
- '−' is binary operation on Z $a \in Z, b \in Z$ then $a-b \in Z$
- '*' is binary operation on N $a \in N, b \in N$ $a * b \in N$

(2) **Algebraic Structure**
A non empty set G equipped with one or more binary operation is called algebraic structure.

- $(G, +, *)$ is an algebraic structure
- $(Z, -, *)$ is an algebraic structure

(3) **Semi Group :** An algebraic structure $(G, *)$ is called semi group if the binary operation $*$ is associative in G. $(a*b)*c = a*(b*c)$ $\forall$ $a, b, c \in G$.

(4) **Monoid :** Let $(G, *)$ be a semi group
If the operation also has an identity element then G is called monoid.

(5) Group : let G be any non empty set equipped with a binary operation denoted by $*$. Then $(G, *)$ algebraic structure is a group if it satisfies the following condition.

$a, b, c \in G$

1. Closure property : $a \in G, b \in G$, $a * b \in G$
2. Associative property : $(a*b)*c = a*(b*c)$
3. Existance of Identity : $\exists$ an identity element $e \in G$ such that $ea = ae = a$
4. Existence of Inverse : $\exists$ an element $a^{-1} \in G$ such that $a * a^{-1} = e$

(commutative group)

(6) Abelian group : A group $(G,*)$ in which binary operation $*$ is commutative is called an abelian group.

Ex : Show that the set of all positive rational numbers forms an abelian group under the composition defined by $a * b = \frac{ab}{2}$

$(Q^+, *)$ $\quad a * b = \frac{ab}{2}$

- let $a, b, c \in Q^+$

1) closure prop :

$$a * b = \frac{ab}{2} \in Q^+$$

2) Associative prop

$$(a*b)*c = \frac{ab}{2} * c = \frac{abc}{4}$$

$$(a*(b*c)) = a * \left(\frac{bc}{2}\right) = \frac{abc}{4}$$

(3) Existence of Identity

we have to find $e \in Q^+$

such that

$$a * e = a$$

$$\frac{ae}{2} = a \qquad (a \neq 0)$$

$$\therefore e = 2 \in Q^+$$

(4) Existence of Inverse :

we have find $a^{-1} \in Q^+$ such that

$$a * a^{-1} = e = 2$$

$$a * a^{-1} = 2$$

$$\frac{aa^{-1}}{2} = 2$$

$$aa^{-1} = 4$$

$$\therefore a^{-1} = \frac{4}{a} \in Q^+$$

(5) Commutative

$$a * b = \frac{ab}{2}$$

$$b * a = \frac{ba}{2}$$

$\therefore a * b = b * a \quad (\because a, b \in Q^+)$

$\therefore (Q^+, *)$ is an abelian group

$\therefore (Q^+, *)$ is a group.

Ex: If $G = \left\{ \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} ; \text{ a is any non-zero real number} \right\}$.

Show that G is a commutative group with matrix multiplication.

Soln: Let $A, B, C \in G$

$$A = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} b & 0 \\ 0 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} c & 0 \\ 0 & 0 \end{bmatrix}$$

(1) Closure property

$$AB = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} b & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} ab+0 & 0+0 \\ 0+0 & 0+0 \end{bmatrix} = \begin{bmatrix} ab & 0 \\ 0 & 0 \end{bmatrix} \in G$$

(2) Associative property

$$(AB)C = \begin{bmatrix} ab & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} c & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} abc & 0 \\ 0 & 0 \end{bmatrix}$$

$$A(BC) = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} bc & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} abc & 0 \\ 0 & 0 \end{bmatrix}$$

$\therefore (AB)C = A(BC)$

(3) Existence of Identity :

let $I = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \in G$

$$A \cdot I = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} = A$$

(4) Existence of inverse :

We have to find $A^{-1} \in G \ \exists$

$AA^{-1} = I$

$$\begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} ae_1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$\therefore ae_1 = 1$

$\therefore e_1 = \frac{1}{a}$

$\therefore A^{-1} \begin{bmatrix} 1/a & 0 \\ 0 & 0 \end{bmatrix}$

$\therefore$ G is a group under matrix multiplication.

- Commutative property

$$AB = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} b & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} ab & 0 \\ 0 & 0 \end{bmatrix}$$

$$BA = \begin{bmatrix} b & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} ab & 0 \\ 0 & 0 \end{bmatrix}$$

$\therefore$ G is an abelian group with matrix multiplication.

Ex : Show that the operation $*$ defined by $x * y = x^y$ on the set N is neither commutative nor associative.

Soln: Let $x = 1, \ y = 2, \ z = 3 \in N$

$x * y = x^y$ $\qquad$ $y * x = y^x$

$1 * 2 = 1^2 = 1$ $\qquad$ $2 * 1 = 2^1 = 2$

$\therefore x * y \neq y * x$

(2) Associative property

$\forall x, y, z \in N$

$\therefore (x * y) * z = x * (y * z)$

$(x * y) * z = (x^y) * z = (x^y)^z$

$x * (y * z) = x * (y^z) = (x)^{y^z}$

$\therefore (x * y) * z \neq x * (y * z)$

Let $x = 2, y = 3, z = 4$

$(x^y)^z = (2^3)^4 = 8^4 = 4096$

$(x)^{y^z} = (2)^{3^4} = (2)^{81}$

$= 2.4179 \times 10^{2}$

Semigroup — closure, Associative

↓

monoid — Semigroup + Identity

↓

Group — monoid + Inverse

↓

Abelian group — group + commutative

Ex: Show that $G = \{a + b\sqrt{2},\ a, b \in Q\}$ is a group with addition

Sol^n: let $x = a + b\sqrt{2},\ y = c + d\sqrt{2},\ z = f + g\sqrt{2} \in G$

(1) closure prop.

$x + y = (a + b\sqrt{2}) + (c + d\sqrt{2})$

$= (a + c) + (b + d)\sqrt{2} \in G$

(2) Associative

$(x+y)+z = [(a+c)+(b+d)\sqrt{2}] + (f+g\sqrt{2})$

$= (a+c+f)+(b+d+g)\sqrt{2}$

$= (a+(c+f))+b\sqrt{2}+(d+g)\sqrt{2}$

$= x+(y+z)$

(3) Identify

since $0 \in Q$ $\therefore e = 0+0\sqrt{2} \in G$

$\therefore x+e = (a+b\sqrt{2})+(0+0\sqrt{2}) = x$

(4) Inverse

since $a, b \in Q$, $-a, -b \in Q$

$\therefore I = -a-b\sqrt{2} \in G$

$\therefore x+I = (a+b\sqrt{2})+(-a-b\sqrt{2}) = 0+0\sqrt{2} = e$

$\therefore G$ is a group

Ex: Show that the algebraic structure $(G, *)$ is a group where

$G = \{(a,b) \mid a, b \in R, a \neq 0\}$

and $*$ is defined as

$(a,b)*(c,d) = (ac, bc+d)$

$\forall (a,b), (c,d) \in G$

Solⁿ: Let $(a,b), (c,d), (f,g) \in G$

with $a \neq 0, c \neq 0, f \neq 0$

(1) closure

$(a,b)*(c,d) = (ac, bc+d) \in G$

(2) Associative

$x=(a,b)$, $y=(c,d)$, $z=(f,g)$

$(x*y)*z = [(a,b)*(c,d)]*(f,g)$

$= (ac, bc+d)*(f,g)$

$= (acf, bcf+df+g)$

$x*(y*z) = (a,b)*[(c,d)*(f,g)]$

$= (a,b)*[cf, df+g]$

$= (acf, bcf, df+g)$

$\therefore (x*y)*z = x*(y*z)$

(3) Identity

We have to find $e=(e_1,e_2) \in G$
Such that $x*e=x$

$(a,b)*(e_1,e_2)=(a,b)$

$\Rightarrow (ae_1, be_1+e_2)=(a,b)$

$\Rightarrow ae_1 = a$, $be_1+e_2=b$

$\Rightarrow e_1 = 1$ $\therefore b+e_2=b$

$\therefore e_2=0$

$\therefore e=(e_1,e_2)=(1,0)\in G$

(4) Inverse

We have to find $I=(a^{-1},b^{-1})\in G$
Such that $x*I=e$

$(a,b)*(a^{-1},b^{-1})=(1,0)$

$(aa^{-1}, ba^{-1}+b^{-1})=(1,0)$

$aa^{-1}=1$, $ba^{-1}+b^{-1}=0$ $\Rightarrow \frac{b}{a}+b^{-1}=0$ $\therefore b^{-1}=-b/a$

$a^{-1}=1/a$

$$I = (a^{-1}, b^{-1})$$
$$= \left(\frac{1}{a}, \frac{-b}{a}\right) \in G$$

$\therefore$ G is a group

(7) Subgroup:

If $\{G, *\}$ is a group and $H \subseteq G$ is a non empty subset, that satisfies the following conditions

1) for $a, b \in H$ $\quad a * b \in H$
2) $e \in H$ where e is the indentity of $(G, *)$
3) for any $a \in H$, $a^{-1} \in H$

then $\{H, *\}$ is a subgraph of $\{G, *\}$

OR,, If $\{G, *\}$ is a group and $H \subseteq G$ is a non empty subset then H is called subgrap if H itself is a group under the operation which is defined for G.

(8) Right coset / lef coset

If $\{H, *\}$ is a subgroup of $\{G, *\}$ then set when $a \in G$

defined by $\quad aH = \{a * h \mid h \in H\}$

is called left coset of G

$Ha = \{h * a \mid h \in H\}$ - Right coset of G

9) Normal Subgroup

A subgroup $\{H, *\}$ of a group $\{G, *\}$ is called normal subgroup, if any $a \in G$

$$aH = Ha \quad \text{or} \quad a * H = H * a$$

(10) Quotient group

If H is a normal subgroup of $\{G, *\}$ and G/H denotes the set of all coset of H in G of the binary operation $\otimes$ is defined as for all $a, b \in G$

$aH \otimes bH = (a * b)H$

then $\{G/H, \otimes\}$ is a Quotient group.

Thm: Lagrange's theorem

The order of each subgroups of a finite group is devision of the order of the group.

$\Rightarrow \quad \frac{O(G)}{O(H)} = k$

(11) congruence relation

Let m be the +ve integer we say that 'a' congruent to b modulo m

$a \equiv b \pmod{m}$

Note: if m divides the diff $a-b$

$a-b = mk$

Note: An equivalence relation on a set is a rule such that

(i) $a \sim a \quad \forall\, a \in S$ (reflexive)

(ii) $a \sim b$ then $b \sim a$ (symmetry)

(iii) $a \sim b$, $b \sim c$, then $a \sim c$ (transitive)

Ex : Show that the relation $x \equiv y \pmod m$ define on the set of integer I is an equivalance relation.

Sol^n: $x \equiv y \pmod m$ $\therefore x - y = mk$ for some $m, k \in Z$

(1) Reflexive

Since $x - x = 0$

$x \equiv x \pmod m$

(2) Symmetry

$x \equiv y \pmod m$

$\Rightarrow$ $x - y = mk$

$y - x = -mk = m(-k) = mk_1$

$y \equiv x$ (mod m

(3) Transitive

$x \equiv y \pmod m \Rightarrow x - y = mk_1$

$y \equiv z \pmod m \Rightarrow y - z = mk_2$

$\therefore x - y + y - z = mk_1 + mk_2$

$x - z = m(k_1 + k_2)$

$x - z = mk_3$

$\therefore x \equiv z \pmod m$

(આગળ નાં પેઇજ માં solve કરેલ છે.)

✓ Ex. Prove that Every cyclic group is an abelian group.

(12) - cyclic group : A group $\{G, *\}$ is said to be cyclic if there exists an element $a \in G$ such that every element $x \in G$ can be expressed as $x = a^n$ for some integer n.

→ a is generator of G.

Ex. $G = \{1, -1, i, -i\}$

$(i)^4 = 1$
$(i)^2 = -1$
$(i)^1 = i$
$(i)^3 = -i$

$\therefore$ i is the generate of the group.

Ex. let $\{G, *\}$ be cyclic group
let 'a' be the generator of G

let $p, q \in G$
$\therefore p = a^m, \; q = a^n$ for same m, n

$p * q = a^m a^n = a^{m+n} = a^n a^m = q * p$

$\therefore$ Every cyclic group is an abelian group.

(13) Permutation
let S be any set. Any bijective mapping of S to S is called a permutation S.

→ $S = \{a, b\}$
$P_1 = \begin{pmatrix} a & b \\ a & b \end{pmatrix}$, $P_2 = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$

Note: (i) let S be any set of n elements $\{1, 2, \ldots, n\}$ then any one to one onto function $f: S \to S$ is called permutation on n symbol.

→ The no. of elements in S is known as degree of permutation.

2) If a set S has n elements then we have n! permutation of S.

3) The set of all permutation on n symbol is denoted by S_n.

4) $\langle S_n, \diamond \rangle$ is group

5) A binary operation $\diamond$ on S_n as follows for any $f, g, \in S_n$

$$f \diamond g = g o f$$

6) Every permutation can be written as product of finite number of transposition.

(14) Transposition : Any cycle of length two is called transposition.

Ex. If $A = (1\ 2\ 3\ 4\ 5)$, $B = (2, 3)\ (4\ 5)$ then find AB

Sol^n $A = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 1 \end{pmatrix}$ $B = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 2 & 5 & 4 \end{pmatrix}$

$AB = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 5 & 4 & 1 \end{pmatrix}$

Ex. If $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{pmatrix}$ $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 5 & 4 \end{pmatrix}$

$\gamma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 1 & 2 \end{pmatrix}$ $\delta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 1 & 5 & 4 \end{pmatrix}$

then find $\alpha\beta$, $\beta\gamma$, $\gamma\delta$, α^{-1}, β^{-1}, γ^{-1}, δ^{-1}
also solve the eqn $\alpha x = \beta$

Soln: $\alpha\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$

$\beta\gamma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 2 & 1 \end{pmatrix}$

$\gamma\delta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 1 & 3 & 2 \end{pmatrix}$

$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{pmatrix} = \alpha^{-1}$

$\alpha^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 1 & 2 & 4 & 5 \end{pmatrix}$

$\beta^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 5 & 4 \end{pmatrix}$

$\delta^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 1 & 5 & 4 \end{pmatrix}$

$\alpha x = \beta \Rightarrow \alpha^{-1}\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 1 & 2 & 4 & 5 \end{pmatrix}\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 5 & 4 \end{pmatrix}$

$x = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 1 & 2 & 5 & 4 \end{pmatrix}$

Ex. Prove that $\langle S_3, \diamond \rangle$ is a non abelian group.

Soln: $S_3 = \{P_1, P_2, P_3, P_4, P_5, P_6\}$

$P_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$

$P_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} = (1\ 2)$

$P_3 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} = (1\ 3)$

$P_4 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} = (2\ 3)$

$P_5 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$

$P_6 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$

◇	P_1	P_2	P_3	P_4	P_5	P_6
P_1	P_1	P_2	P_3	P_4	P_5	P_6
P_2	P_2	P_1	P_5	P_6	P_3	P_4
P_3	P_3	P_6	P_1	P_5	P_4	P_2
P_4	P_4	P_5	P_6	P_1	P_2	P_3
P_5	P_5	P_4	P_2	P_3	P_6	P_1
P_6	P_6	P_3	P_4	P_2	P_1	P_5

1) closure property

$P_i \diamond P_j \in S_3$
$\forall\, i, j = 1, 2, \ldots, 6$

2) Associative property

$P_i \diamond (P_j \diamond P_K) = (P_i \diamond P_j) \diamond P_K$
$\forall\, i, j, k = 1, 2, \ldots, 6$
we know that composition function on a binary operation is always associative

3) Identity element
Clearly from the table
$P_1 \diamond P_j = P_j \quad \forall j = 1, 2, \ldots 6$
$\therefore$ P_1 is the identity element.

4) Inverse :

From the table

Inverse of P_1 is P_1

P_2 is P_2

P_3 is P_3

P_4 is P_4

P_5 is P_6

P_6 is P_5

$\therefore \langle S_3, \diamond \rangle$ is a group

- From the table $P_2 \diamond P_3 = P_5$

$P_3 \diamond P_2 = P_6$

$\therefore \langle S_3, \diamond \rangle$ is a non abelian group.

Ex : If $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 5 & 4 & 6 & 2 \end{pmatrix}$ and $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 3 & 1 & 6 & 2 & 4 \end{pmatrix}$

find $\alpha\beta$, $\beta\alpha$, α^2, β^2, α^{-1} and β^{-1}

Soln: $\alpha\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 5 & 4 & 6 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 3 & 1 & 6 & 2 & 4 \end{pmatrix}$

$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 5 & 2 & 6 & 4 & 3 \end{pmatrix}$

$\beta\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 3 & 1 & 6 & 2 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 5 & 4 & 6 & 2 \end{pmatrix}$

$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 3 & 2 & 1 & 4 \end{pmatrix}$

$\alpha^2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 5 & 4 & 6 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 5 & 4 & 6 & 2 \end{pmatrix}$

$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 3 & 6 & 4 & 2 & 1 \end{pmatrix}$

$$\beta^2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 3 & 1 & 6 & 2 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 3 & 1 & 6 & 2 & 4 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 5 & 4 & 3 & 6 \end{pmatrix}$$

$$\alpha^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 6 & 1 & 4 & 3 & 5 \end{pmatrix}$$

$$\beta^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 5 & 2 & 6 & 1 & 4 \end{pmatrix}$$

(15) Homomorphism

If $\{G, *\}$ and $\{G', \Delta\}$ are two groups then a mapping $f: G \to G'$ is called group homomorphism if for any $a, b \in G$

$$f(a * b) = f(a) \,\Delta\, f(b)$$

(16) Isomorphism

A group homomorphism is said to be isomorphism if f is one one & on to.

(17) Ring :

An algebraic system $(G, +, \cdot)$ where G is any non empty set is called ring if follow coditions are satisfied.

1) $(G, +)$ is an abelian group.
2) $(G, \cdot)$ is a semi group.
3) The operation $\cdot$ is distributive over $+$

for any $a, b, c \in R$

$$a \cdot (b + c) = a \cdot b + a \cdot c$$
$$(b + c) \cdot a = b \cdot a + c \cdot a$$

Ex: Show that $\{ Z, \oplus, \odot \}$ is commutative ring with identity where addition and multiplication is defined as

$$\left.\begin{aligned} a \oplus b &= a+b-1 \\ a \odot b &= a+b-ab \end{aligned}\right\} \forall a, b \in Z$$

Sol^n: let $a, b, c, d \in Z$

1) Closure

$a \oplus b = a+b-1 \in Z$

2) Associative

$(a \oplus b) \oplus c = (a+b-1) \oplus c = a+b-1+c-1$
$= a+b+c-2$

$(a \oplus (b \oplus c)) = a \oplus (b+c-1) = a+b+c-1 = a+b+c-2$

$\therefore$

3) Identity

we have to find $e \in Z \ni a \oplus e = a$

$a+e-1 = a$

$e-1 = 0$

$e = 1 \in Z$

4) Inverse

we have to find $a^{-1} \in Z \ni a \oplus a^{-1} = e = 1$

$a + a^{-1} - 1 = 1$

$a + a^{-1} = 2$

$a^{-1} = 2 - a \in Z$

5) Commutative

$a \oplus b = a+b-1$ $\therefore (Z, \oplus)$ is an abelian group

$= b+a-1$

$= b \oplus a$

6) closure

$a \odot b = a+b-ab \in Z$

7) Associative

$(a \odot b) \odot c = (a+b-ab) \odot c = a+b+c-ab-ac-bc+abc$

$a \odot (b \odot c) = a \odot (b+c-bc) = a+b+c-bc-ab-ac+abc$

8) $a.(b+c) = a.(b+c-1)$

$= a+b+c-1-ab-ac+a$

$= \underline{a+b-ab} + \underline{a+c-ac} \; \underline{-1}$

$= a.b + a.c$

$(b+c).a = (b+c-1).a$

$= b+c-1+a-ab-ac$

$= a+b-ab+a+c-ac-1$

$= b.a + c.a$

(18) without zero divisor

If a and b are two elements of G then G is without zero divisor if $a.b = 0 \Rightarrow a = 0$ or $b = 0$

or $a \neq 0 \;\; b \neq 0 \Rightarrow ab \neq 0$

(19) with zero divisor

If $a \neq 0 \;\; b \neq 0 \Rightarrow ab = 0$ then the set is known as with zero divisor.

(20) Integral domain

- A commutative ring without zero divisor is an Integral domain.

(21) Field

- A commutative ring with unity in which every non-zero element possesses their multiplicative inverse.

i.e. $(F, +, \cdot)$ is a field of

1) $(F, +)$ is an abelian group
2) $(F, \cdot)$ is an abelian group
3) Multiplication is distributive over addition.

Ex: If M is the set of 2×2 matrices of the form $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$ where $a, b \in Z$
is the set of integers where $\oplus$ and $\odot$ represent matrix addition and matrix multiplication show that the mapping $f: M \to Z$ given by $f\left(\begin{bmatrix} a & b \\ b & a \end{bmatrix}\right) = b - a$ is a

homomorphism show that $(M, \oplus, \odot)$ is a (ring) group.

Soln: Let $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ $B = \begin{bmatrix} c & d \\ d & c \end{bmatrix}$, $E = \begin{bmatrix} g & h \\ h & g \end{bmatrix} \in M$

- first we prove that $(M, \oplus)$ is a commutative (ring), group

1) Closure

$$A + B = \begin{bmatrix} a & b \\ b & a \end{bmatrix} + \begin{bmatrix} c & d \\ d & c \end{bmatrix} = \begin{bmatrix} a+c & b+d \\ b+d & a+c \end{bmatrix} \in M$$

2) Associative

$$(A+B)+C = \begin{bmatrix} a+c & b+d \\ b+d & a+c \end{bmatrix} + \begin{bmatrix} g & h \\ h & g \end{bmatrix}$$

$$= \begin{bmatrix} a+c+g & b+d+h \\ b+d+h & a+c+g \end{bmatrix}$$

$$= \begin{bmatrix} a & b \\ b & a \end{bmatrix} + \begin{bmatrix} c+g & d+h \\ d+h & c+g \end{bmatrix}$$

$$= A+(B+C)$$

3) Additive Identity :

Since the addition is usual matrix addition

$\therefore E = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \in M \ni A+E = \begin{bmatrix} a & b \\ b & a \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$$= \begin{bmatrix} a & b \\ b & a \end{bmatrix} = A$$

(4) Additiv Inverse

Since $a, b \in Z$ then $-a, -b \in Z$.

$$A = \begin{bmatrix} a & b \\ b & a \end{bmatrix} \Rightarrow A^{-1} = \begin{bmatrix} -a & -b \\ -b & -a \end{bmatrix} \in M$$

$$A+A^{-1} = \begin{bmatrix} a & b \\ b & a \end{bmatrix} + \begin{bmatrix} -a & -b \\ -b & -a \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = E$$

(5) Commutative

$$A+B = \begin{bmatrix} a & b \\ b & a \end{bmatrix} + \begin{bmatrix} c & d \\ d & c \end{bmatrix} = \begin{bmatrix} a+c & b+d \\ b+d & a+c \end{bmatrix} = \begin{bmatrix} c+a & d+b \\ d+b & c+a \end{bmatrix}$$

$= B+A$ $\quad (\because a,b,c,d \in z)$

$\therefore (M, \oplus)$ is a commutative group.

→ Now, we prove that $(M, \odot)$ is a semi group.

(6) Closure

$$A \cdot B = \begin{bmatrix} a & b \\ b & a \end{bmatrix} \cdot \begin{bmatrix} c & d \\ d & c \end{bmatrix}$$

$$= \begin{bmatrix} ac+bd & ad+bc \\ bc+ad & bd+ac \end{bmatrix} \in M$$

(7) Associative

$$(A \cdot B) \cdot C = \begin{bmatrix} ac+bd & ad+bc \\ ad+bc & ac+bd \end{bmatrix} \begin{bmatrix} g & h \\ h & g \end{bmatrix}$$

$$= \begin{bmatrix} acg+bdg+adh+bch & ach+bdh+adg+bcg \\ adg+bcg+ach+bdh & adh+bch+acg+bdg \end{bmatrix}$$

$$= \begin{bmatrix} a & b \\ b & a \end{bmatrix} \cdot \begin{bmatrix} cg+dh & ch+dg \\ ch+dg & cg+dh \end{bmatrix}$$

$$= A \cdot (B \cdot C)$$

- $(M, \odot)$ is a semigroup.

(8) Distributive

$$A \cdot (B+C) = \begin{bmatrix} a & b \\ b & a \end{bmatrix} \cdot \begin{bmatrix} c+g & d+h \\ d+h & c+g \end{bmatrix}$$

$$= \begin{bmatrix} ac+ag+bd+bh & ad+ah+bc+bg \\ bc+bg+ad+ah & bd+bh+ac+ag \end{bmatrix}$$

$$= \begin{bmatrix} ac+bd & ad+bc \\ ad+bc & ac+bd \end{bmatrix} + \begin{bmatrix} ag+bh & ah+bg \\ bg+ah & bh+ag \end{bmatrix}$$

$$= A \cdot B + A \cdot C$$

∴ Similary $(B+C) \cdot A = B \cdot A + C \cdot A$

∴ $(M, \oplus, \odot)$ is a commutative Ring.

→ To prove f is a homomorphism

let $M_1 = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ & $M_2 = \begin{bmatrix} c & d \\ d & c \end{bmatrix} \in M$

$$M_1 + M_2 = \begin{bmatrix} a & b \\ b & a \end{bmatrix} + \begin{bmatrix} c & d \\ d & c \end{bmatrix} = \begin{bmatrix} a+c & b+d \\ b+d & a+c \end{bmatrix}$$

$$f(M_1 + M_2) = (b+d) - (a+c)$$
$$= (b-a) + (d-c)$$
$$= f(M_1) + f(M_2)$$

Ex: Construct the composition tables for
(i) addition modulo 4 and
(ii) multiplication modulo 4 for
$Z_4 = \{0, 1, 2, 3\}$. Check whether they
have identity and inverse element
{GTU 2020}

Sol^n: $Z_4 = \{[0], [1], [2], [3]\} = \{0, 1, 2, 3\}$
elements of Z_4 are equivalence classes
generated by 0, 1, 2, 3

$+_4$ $[a] +_4 [b] = [(a+b) \pmod 4]$
= [Remainder when a+b is divided by 4]

$\times_4$ $[a] \times_4 [b]$ = [Remainder when ab is divided by 4]

$3 +_4 2 = 4\overline{)5}$ → quotient 1, $5 - 4 = 1$ $\quad 3 +_4 2 = 1$

$[3] +_4 [2] = [1]$

composition modulo 4 table for addition

$+_4$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

$0 +_4 0 = 0$ $\quad 4\overline{)0}$ → quotient 0, remainder 0

$0 +_4 1 = 1$ $\quad 4\overline{)1}$ → quotient 0, remainder 1

$a < 4$ then answer element itself

$a +_4 e = a = e +_4 a$
for all a

→ clearly from the table [0] is the identity for addition modulo 4

Inverse $0 +_4 \underline{0} = 0$ ∴ Inverse of 0 is 0

$1 +_4 \underline{3} = 0$ ∴ Inverse of 1 is 3

$2 +_4 \underline{2} = 0$ ∴ Inverse of 2 is 2

$3 +_4 \underline{1} = 0$ ∴ Inverse of 3 is 1

→ Composition table for multiplication Modulo 4:

$\times_4$	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

$a \times_4 b$ = Remainder

$0 \times_4 1 = 0$

$1 \times_4 1 = 1$

$2 \times_4 1 = 2$

$3 \times_4 1 = 3$

So [1] is the identity element

→ $0 \times_4 a = 1$ no inverse for 0

$1 \times_4 1 = 1$ ∴ 1 is self inverse

$2 \times_4 2 = 1$ ∴ 2 has no inverse

$3 \times_4 3 = 1$ ∴ 3 is self inverse

→ From the table we find that

$a \times_4 1 = 1 \times_4 a = a$ for $a = 0, 1, 2, 3$

∴ 1 is the identity in Z_4 with respect to multiplication modulo 4.

→ From the table, we observe that

$1 \times_4 1 = 1$ and $3 \times_4 3 = 1$

∴ 1 and 3 are self inverses with respect to $\times_4$

→ Also from the table, we observe that 0 and 2 has no inverse.